Research Experience: MPI-P

Asymmetry & Anti-correlation of Hydrogen Bonds in liquid water

MAX PLANCK INSTITUTE
FOR POLYMER RESEARCH

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- **Organization**

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- **Type**

Full-time Research Internship

- **Period**

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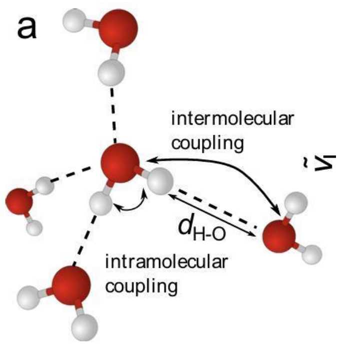
Background & Goals

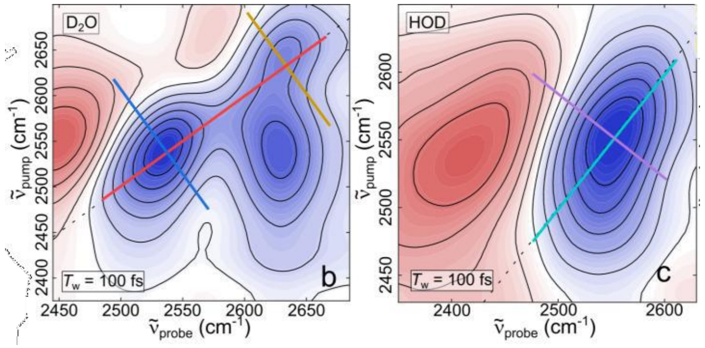
Hydrogen Bond Network

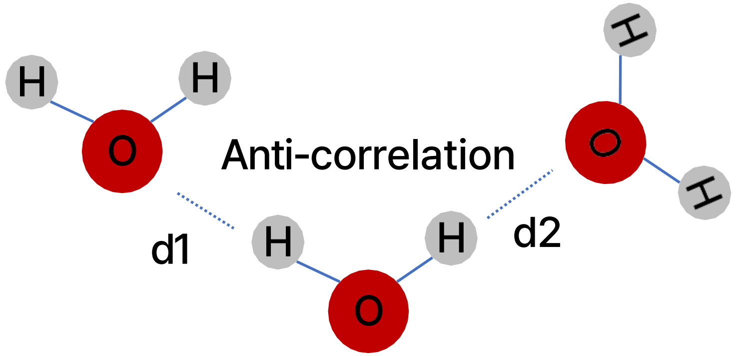


- Water has anomalous physical properties attributed to its hydrogen-bond network.
- However, the conventional tetrahedral picture is insufficient to describe its local structural heterogeneity.
- Two O–H bonds in a single molecule can form unequal hydrogen bonds, leading to structural asymmetry.

Anti-correlation of Liquid Water

Our group observed anti-correlation between two O–H stretch modes using 2D-IR, suggesting intramolecular asymmetry





Yet, their origin and structural distribution remain unclear.

Research Questions

Q1. What is the origin of hydrogen-bond asymmetry or anti-correlation in water? (Thermal Factor?)

Q2. Can we prove anti-correlation in simulation?

Specifically:

- How does temperature affect hydrogen–bond heterogeneity?
- Can anti–correlation be reproduced and quantified in simulations?
- What is the relationship between asymmetry and anti–correlation?

Approach

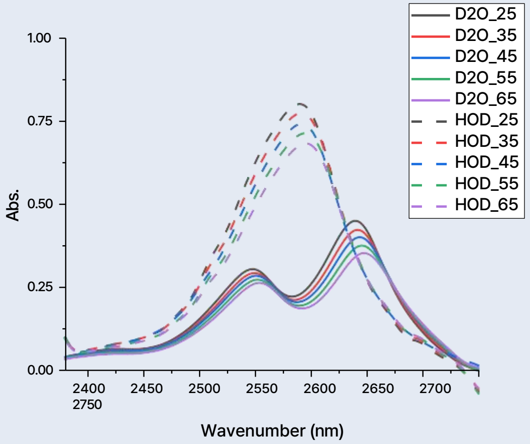
Combined temperature-dependent FTIR spectroscopy with molecular simulations to probe hydrogen-bond network structures and vibrational behavior.

Experiment – FTIR

- Temperature-dependent FTIR measurements (25–65 °C)
- HOD and D_2O diluted in DMF were used to:

  -suppress intermolecular coupling

  -isolate intramolecular vibrational behavior
- The O–D stretching band (~2500 cm^{-1}) was analyzed.



Simulation – MD simulation

- Classical molecular dynamics simulations to model water structure.
- Two water models were used:

  -SPC/E (non-polarizable)

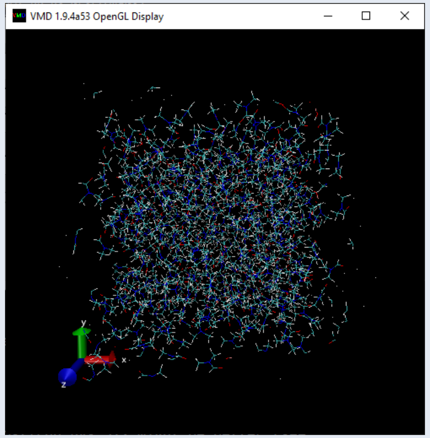
  -SWM4-NDP (polarizable)
- Analysis:

  -# of HB

  -HB distances (d_1,d_2)

  -Radial distribution function

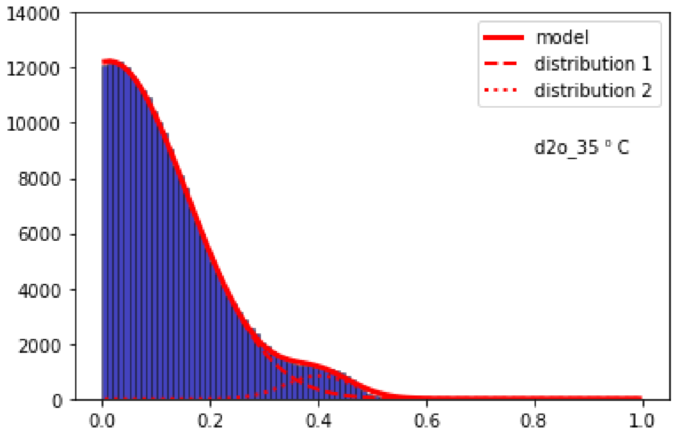
  -Time correlation function and VDOS

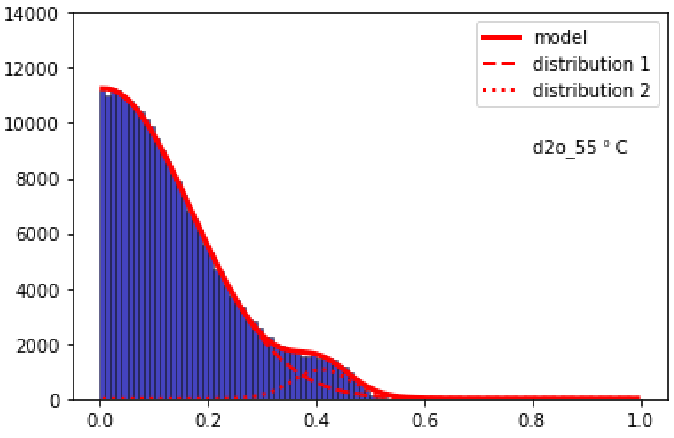


Results

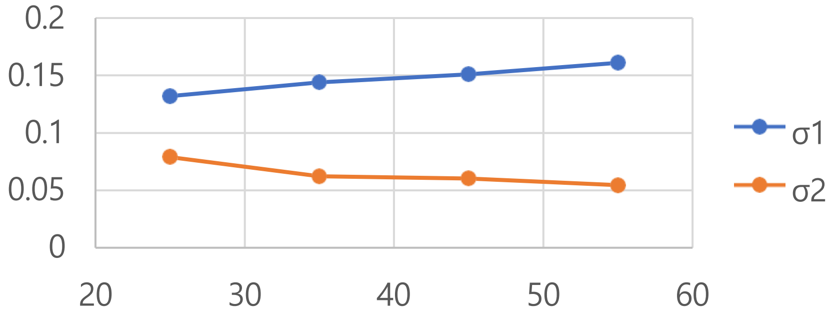
Asymmetry Distribution

Define Asymmetry Parameter $\gamma = 1 - \frac{\min(d1,d2)}{\max(d1,d2)}$





Gaussian Fit with $f = A\exp(-\frac{(x-\mu)^2}{2\sigma^2})$



- Additional distinct high-asymmetry regime emerges.
- Bimodal Gaussian fit show that the main distribution broadens with temperature, while the high-asymmetry component becomes narrower but increases in contribution.

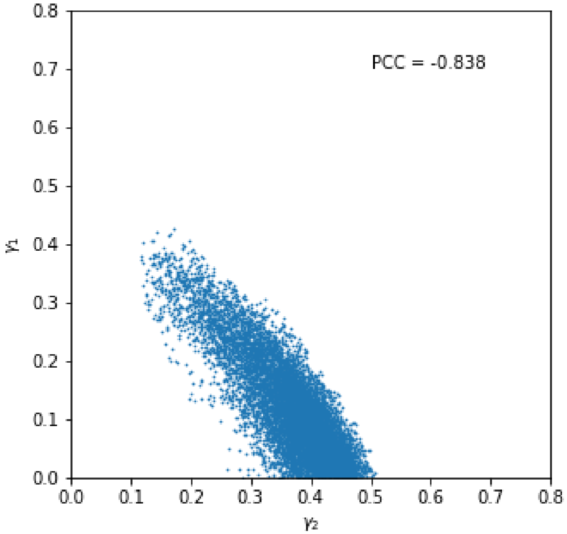
➔ Increasing thermal energy does not only induce random fluctuations, but also enhances the population of a structurally distinct asymmetric hydrogen-bonding state.

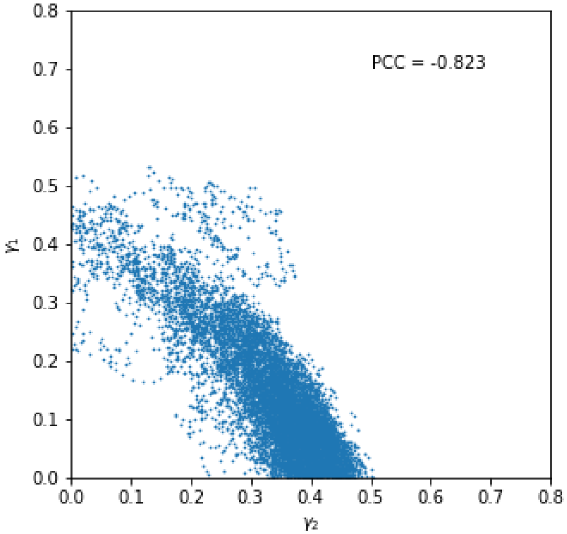
- Based on this observation, let's further investigated its possible connection to the anti-correlation behavior.

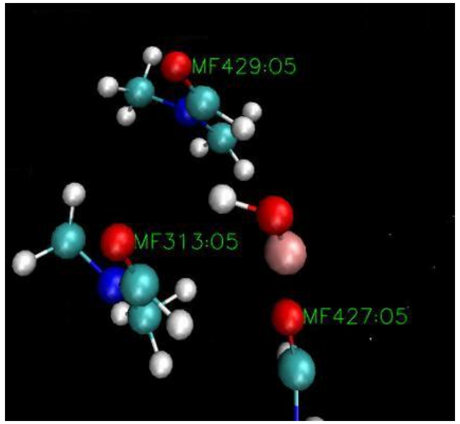
Results

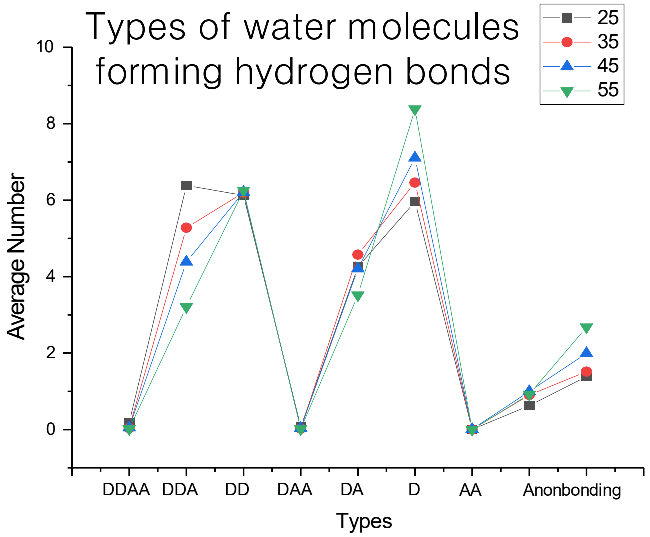
Non-bound Water molecules

Define ɣ2 : asymmetry parameter of nearest and 2nd nearest interaction distances of the less bound hydrogen







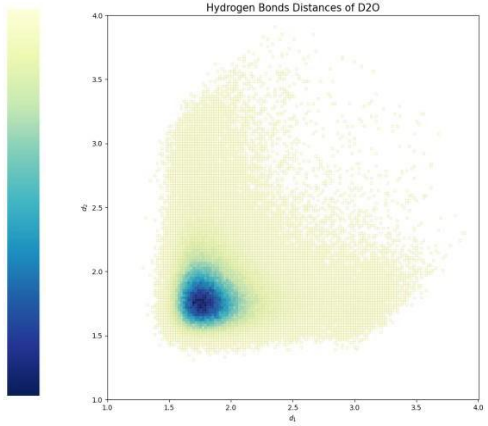


- Significant negative correlation between ɣ1 and ɣ2.
- Appearance of distinctly asymmetric state attributed to competing interaction, which makes "non-bound water" of "non-bonding OH group."
- Temperature ↑ → # of non-bound water molecules ↑.
- Two HB distances have positive correlation when they are highly asymmetric.

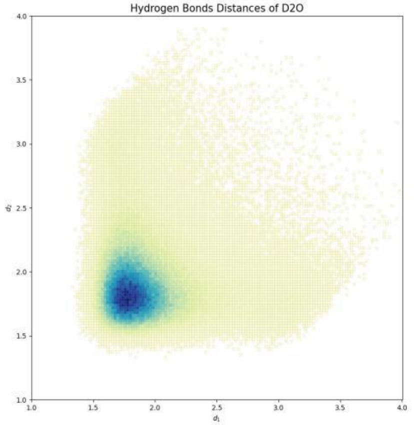
Results

Anti-correlation

HB distances at 25 ˚C

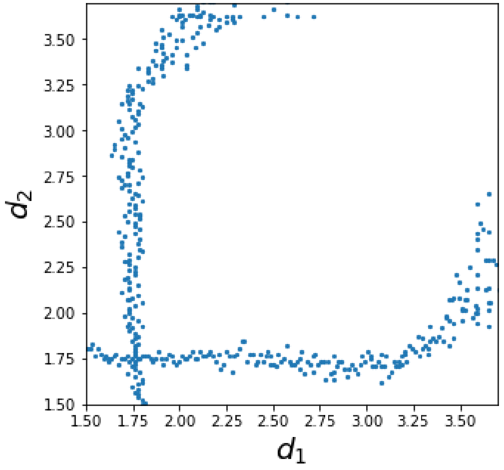


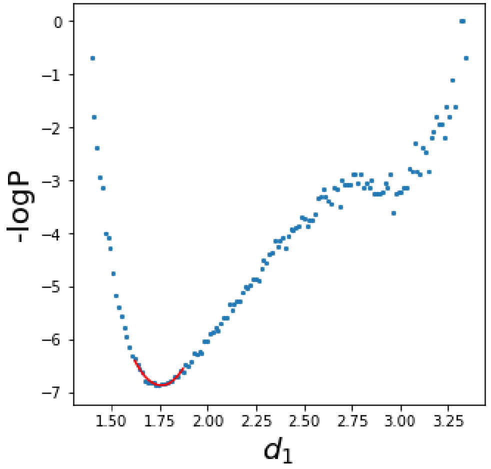
HB distances at 55 ˚C



- Analyzed the conditional free energy landscape via $F(d_2 \mid d_1) = -\log P(d_2 \mid d_1)$ with the HB distribution as shown.
- Anti–correlation between d_1 and d_2 is observed across all temperatures, while the strength of coupling decreases as temperature increases.

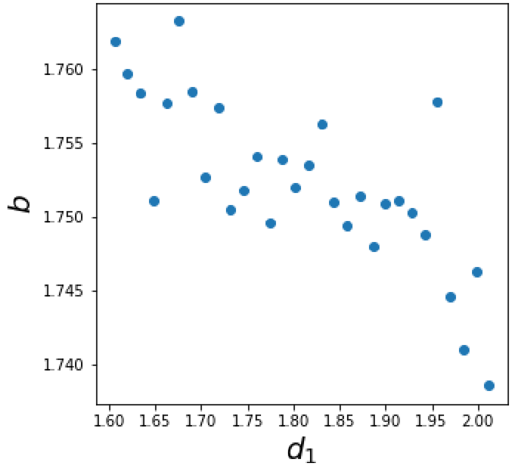
➔ Thermal fluctuations partially weaken the effective coupling, while the anti–correlation is an intrinsic structural feature of liquid water rather than a purely thermal effect.

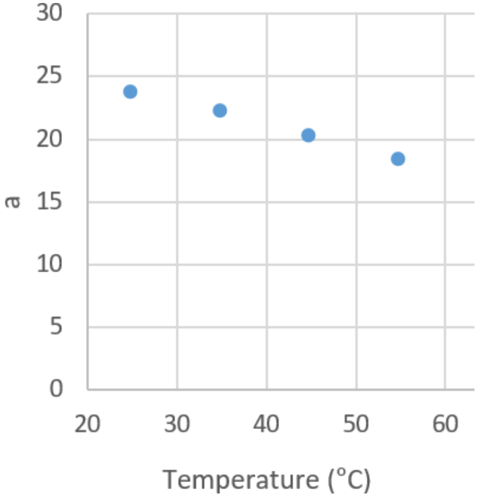




Fit the potential curve around minimum with harmonic approximation

$$F(d1) = a(x - b)^2 + c$$





Conclusion & Insights

- Temperature-dependent FTIR and MD simulations revealed progressive weakening and reorganization of the HB network.
- The anti-correlation behavior was reproduced in MD simulations.
- Anti-correlation is an intrinsic dynamic phenomenon rather than a simple statistical or thermal population effect in contrast to static structural asymmetry.
- Thermal fluctuation promoted the formation of non-bound water populations (which led to the change in vibrational spectrum) and weakened intermolecular vibrational coupling, leading to a more heterogeneous and weekly correlated vibrational ensemble.
- Vibrational responses in water are governed by collective intermolecular coupling dynamics as well as local HB strength.